Qatar-1b

R-1921

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_131118` · created 2026-08-06T00:50:11+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456614.66964 +/- 0.00016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.057
Transit depth (Rp/Rs)^2	3.66 +/- 2.17 [%]
Orbital Inclination (inc)	82.63 +/- 2.88
Airmass coefficient 1 (a1)	0.0137 +/- 0.004
Airmass coefficient 2 (a2)	2.93 +/- 0.8
Scatter in the residuals of the lightcurve fit is	27.86 %
Best Comparison Star	#3 - [588, 20]
Optimal Aperture	2.14
Optimal Annulus	5.35
Transit Duration (day)	0.033 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.057 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.033 d vs 0.0692 d (deviation 2.58σ)
Mid-time O–C	-0.0 min
Depth SNR	3.4
Residual scatter	27.86 %

Light curve

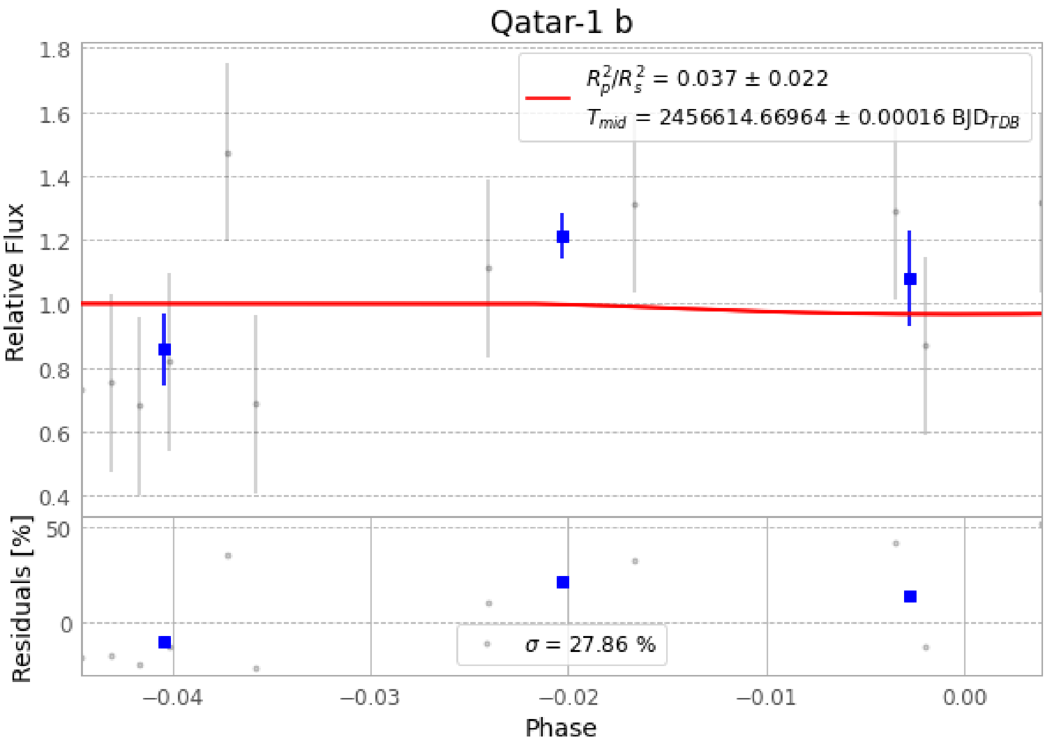

Figure 1 — FinalLightCurve_Qatar-1 b_2013-11-17.png (SHA-256 65f6a9c32c83a3d1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 277], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8eb8cee476fc39ba…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

72 FITS frames (47 MB), `Qatar-1131118023012.FITS` → `Qatar-1131118060312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-131118.log (a29ff667f812…), FinalLightCurve_Qatar-1 b_2013-11-17.png (65f6a9c32c83…), FinalLightCurve_Qatar-1 b_2013-11-17.csv (426f4dfa1ae3…)

Compute resources

Wall time 288.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)